

Gap-Closing Literature Note for the Credit-Card Customer NPV Proposal

What Existing Literatures Can and Cannot Close

Chak Wong

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Abstract

This note is a companion to the credit-card customer NPV proposal. Its purpose is not to restate the proposal or to replace the stochastic-process note. It is to map the proposal's remaining theoretical and modeling gaps to the relevant literatures and to state, gap by gap, what those literatures can actually support. The note is therefore a committee-facing cross-walk. For each major gap, it identifies the exact proposal sections and equations where the issue matters, explains why the gap is material, cites the literature families that can close or tighten it, distinguishes literature support from proposal derivation and bank-specific empirical work, and states what remains unresolved even after the literature is added. The note is intentionally conservative: many gaps can be narrowed substantially by existing literature, but few are eliminated without issuer-specific data, experiments, and governance choices.

1 Introduction and scope

The current proposal has become much stronger on valuation semantics, component decomposition, and dynamic structure. It now contains a long section that assembles the proposal into a Bayesian state-space / belief-state control system. Even so, a committee of econometrics and stochastic-control reviewers can still press on deeper theoretical and modeling gaps. Some are about mathematical structure: is the state sufficient, is the filtration explicit, is the joint kernel coherent? Others are about modeling and identification: how is outside wallet identified, what is the state unit, when is relationship value causal rather than confounded, and where exactly does fixed-policy valuation end and optimization begin?

This note answers those objections in a structured way. It does not claim that literature alone solves the bank's problem. Instead, it uses literature to do four narrower jobs:

1. justify formal language or abstraction,
2. justify a modeling move or computational approximation,
3. show what has already been studied in nearby dynamic decision settings,
4. and mark the line between what can be defended theoretically and what still requires internal bank evidence.

2 How to read this note

Each gap section follows the same template.

1. **Gap in current proposal.** What remains under-specified or vulnerable.
2. **Where the gap appears.** Exact section or equation references in the main proposal.
3. **Literatures that can close or tighten it.** The literature family and key citations.
4. **What the literature actually supports.** The strongest claim that can be defended.
5. **What remains bank-specific.** The part that still must be treated as an issuer-specific empirical object, assumption, or governance rule.

The note uses four support labels throughout:

1. *literature-supported mechanism*,
2. *literature-supported formalism*,
3. *proposal derivation*,
4. *bank-specific empirical object or governance convention*.

A reader should not infer that all gaps can be closed to the same degree. Several can be tightened greatly at the theoretical level while remaining only partially resolved empirically.

3 Crosswalk table

Proposal equations live in the main proposal file, not in this companion note, so all equation anchors below are written as plain labels in parentheses rather than as live `\eqref` cross-references. The purpose is provenance, not internal LaTeX navigation.

Table 1: Gap-to-literature crosswalk for the main proposal.

Gap category	Main proposal anchors	Literature family	What the literature can support	What remains bank-specific
State unit / aggregation	proposal objective (eq:proposal-objective), state kernel (eq:state-kernel-panel), balance stock-flow (eq:balance-stock-flow)	state aggregation, approximate DP, CLV state models	disciplined aggregation and compression of dynamic states	exact bank unit and sufficient aggregation

Gap category	Main proposal anchors	Literature family	What the literature can support	What remains bank-specific
Markov sufficiency / filtration	dynamic-policy system (eq:dynamic-policy-system), policy value (eq:policy-value), state-space assembly section	MDP, DTR, POMDP, Bayesian filtering	explicit information-state and belief-state language	whether the chosen bank features are empirically sufficient
Shared macro common state	category spend (eq:category-spend), PD transition (eq:pd-transition), term/loss mapping (eq:terms-loss)	factor models, macro state-space models, scenario design	low-dimensional common-shock macro block	exact latent macro law and elasticities
Joint kernel / factorization	state kernel (eq:state-kernel-panel), balance stock-flow (eq:balance-stock-flow), PD transition (eq:pd-transition)	dynamic state-space decomposition, modular conditional laws	coherent structured approximation to the joint kernel	exact factorization validity
Belief approximation / compression	stochastic-process section belief-state equations	Bayesian filtering, belief compression, finite-memory approximations	approximate posterior summaries as implementation objects	exact compressed belief sufficient for this bank
Incremental baseline semantics	proposal objective (eq:proposal-objective), NPV functional (eq:panel-npv-functional), policy value (eq:policy-value)	causal inference, DTRs, policy evaluation	fixed-policy value vs optimized-policy value distinction	approved baseline and policy-bundle semantics
Wallet / latent demand identification	wallet accounting through cannibalization-to-NPV (eq:wallet-accounting)- (eq:cannibalization-to-npv)	payment choice, demand/substitution, uplift, partial identification	latent wallet / substitution language and bounded identification	issuer-specific wallet capture and movable fraction

Gap category	Main proposal anchors	Literature family	What the literature can support	What remains bank-specific
Relationship value integration	relationship value (eq:relationship-value), valuation semantics sections	CLV, customer equity, banking profitability	cautious separation of core card value and spillover value	causal spillover magnitude and inclusion rule
Valuation vs optimization boundary	dynamic-policy system (eq:dynamic-policy-system), policy value (eq:policy-value), allowed-use and governance sections	MDP, DTR, OPE, safe improvement, offline RL	fixed-policy simulation first, challenger evaluation second	production optimization threshold and governance gate

4 State unit and aggregation

Gap in current proposal. The proposal allows the unit i to be a customer, account, prospect, household, application, or account-action unit, while the dynamic sections also use wallet, relationship, and all-bank incremental value objects. This creates a structural question: what is the actual dynamic state unit? If the state is too fine, the process becomes intractable; if it is too coarse, internal cannibalization and relationship effects may be misrepresented.

Where the gap appears. This issue enters through proposal objective (eq:proposal-objective), state kernel (eq:state-kernel-panel), wallet accounting (eq:wallet-accounting), balance stock-flow (eq:balance-stock-flow), and the new state-space assembly section where observed and latent state blocks are introduced.

Literatures that can close or tighten it. State aggregation and approximate dynamic programming literature directly helps here. The most useful starting point is Bean et al. [2], which is a general operations-research reference for aggregation in dynamic programming. For a banking or customer-relationship context, CLV and duration/state models already present in the proposal’s bibliography also help define economically meaningful state blocks [6, 11, 28]. Banking-specific dynamic optimization references such as So and Thomas [30] and Lewis [21] can be used to show how dynamic pricing/control papers work with deliberately chosen state abstractions.

What the literature actually supports. The literature supports the claim that state aggregation is a legitimate and often necessary modeling device, and that economically distinct blocks should not be collapsed carelessly if they carry different transition or reward semantics. It also supports the use of lifecycle, balance, payment, risk, and relationship states as meaningful dynamic components.

What remains bank-specific. The literature does not choose the bank’s unique correct state unit. It does not prove whether the correct object is account-level, customer-level, household-level, or a multi-account relationship state. That remains a proposal and governance choice to be defended against the bank’s actual use cases and internal cannibalization structure.

5 Markov sufficiency and filtration

Gap in current proposal. The proposal now distinguishes observed state, latent state, and belief state, but reviewers can still ask whether the chosen information state is truly sufficient. If omitted history matters beyond the proposal’s declared observed state and posterior state, then the Bellman recursion is not properly grounded.

Where the gap appears. This issue appears in the proposal’s dynamic-policy language around dynamic-policy system (eq:dynamic-policy-system) and policy value (eq:policy-value), and more sharply in the full stochastic-process section where the filtration and belief-state objects are introduced.

Literatures that can close or tighten it. This is one of the clearest places where the literature can add theoretical weight. The MDP text of Puterman [25], the dynamic treatment-regime reference Murphy [23], and especially the POMDP / state-space literature [9, 16, 20, 29, 35] provide the correct language for explicit filtration, belief-state sufficiency, and recursive updating.

What the literature actually supports. These references strongly support the need to write down: - a pre-action information set, - a post-observation information set, - a belief or filtering distribution over latent state, - and a policy measurable with respect to the available information. They justify the proposal’s move from static feature stacks to an information- state or belief-state formulation.

What remains bank-specific. The literature does not prove that the proposal’s chosen observed state is actually Markov-sufficient for this bank’s problem. That depends on the bank’s real data lags, omitted state variables, missingness, delayed collections signals, future-policy dependence, and empirical adequacy of the latent-state model.

6 Shared macro common-shock state

Gap in current proposal. The proposal repeatedly uses macro and labor-market variables in spend, payment, loss, and scenario language, but a committee can still ask: how exactly should a common macro state be represented? If every module conditions on macro variables separately, the model risks being a loose collection of reduced-form controls rather than one coherent common-shock dynamic system.

Where the gap appears. This matters in the macro section, especially category spend (eq:category-spend), and in the balance, payment, and loss equations such as PD transition (eq:pd-transition) and term/loss mapping (eq:terms-loss). It also matters in the state-space assembly section’s macro block.

Literatures that can close or tighten it. The most useful gap-closing literatures here are dynamic factor and data-rich macro-state literatures. Bernanke et al. [5] and Bernanke and Boivin [4] support reducing a high-dimensional macro environment to a lower-dimensional latent common state, while Stock and Watson [31] supports the use of dynamic factor structures as inputs to multivariate dynamic systems.

What the literature actually supports. These references support the proposal’s use of a low-dimensional shared macro state or latent factor block that can feed multiple modules jointly—spend, payment, delinquency, funding, and attrition—rather than treating macro state as unrelated controls in each equation.

What remains bank-specific. The literature does not identify the bank’s exact latent macro regime, how many factors to retain, how geography enters, or the issuer-specific elasticities of spend, payment, and risk with respect to that macro state.

7 Joint kernel and factorization

Gap in current proposal. The proposal has many useful module equations, but the committee can still ask whether they define one coherent joint stochastic kernel or merely a set of local conditional blocks that may not compose cleanly.

Where the gap appears. This issue sits behind state kernel (eq:state-kernel-panel), balance stock-flow (eq:balance-stock-flow), payment component (eq:payment-component), line component (eq:line-component), and PD transition (eq:pd-transition), and it becomes sharper in the state-space assembly where routing, balances, payments, delinquency, promotions, and relationship value all appear as blocks in one process.

Literatures that can close or tighten it. The state-space and recursive-filtering literature already in the bibliography is helpful here because it supports writing joint dynamic systems through structured conditional laws [8, 9, 20, 35]. This does not by itself prove the proposal’s factorization, but it supports the idea that modular conditional specifications are an acceptable way to approximate high-dimensional joint dynamics.

What the literature actually supports. The literature supports structured decomposition of a high-dimensional process into conditional blocks when the objective is tractable filtering, simulation, and control. It also supports the idea that exact joint laws are often replaced by modular approximations that must then be validated.

What remains bank-specific. The exact factorization used in the proposal remains a proposal-specific approximation. The literature cannot prove that the bank’s chosen decomposition is exact rather than a pragmatic modeling choice.

8 Belief approximation and compression

Gap in current proposal. Once the proposal introduces latent state and belief-state control, it inherits a tractability problem: posterior beliefs can be very high-dimensional. A committee will

ask how the bank will retain a usable information state without pretending that a low-dimensional approximation is exact.

Where the gap appears. This issue appears in the state-space assembly section’s belief-state equations and filtering discussion, especially where the proposal suggests finite-dimensional posterior summaries.

Literatures that can close or tighten it. The Bayesian filtering and POMDP literatures provide the formal backdrop. In addition, belief-approximation and finite-memory references can tighten the story. The POMDP approximation literature [16, 29] and more explicit aggregation / finite-memory references such as Bean et al. [2] can support the idea that reduced belief summaries may be legitimate if their approximation role is made explicit. The proposal’s new filtering section already cites the core state-space and sequential Monte Carlo references [8, 9, 35].

What the literature actually supports. The literature supports the distinction between the exact sufficient statistic (the full posterior distribution) and an operational approximation used for tractability. It also supports the view that approximate filtering and finite-dimensional compression are legitimate implementation devices.

What remains bank-specific. The exact compressed belief state the bank should use is not identified by the literature. Its quality must be evaluated in the bank’s setting by calibration, policy-value stability, and sensitivity analysis.

9 Incremental baseline semantics in dynamic settings

Gap in current proposal. The proposal is strong on incremental valuation, but dynamic settings make the baseline more subtle. Value depends not only on the current action but also on the downstream policy bundle, the horizon, and the counterfactual path.

Where the gap appears. This gap appears in proposal objective (eq:proposal-objective), NPV functional (eq:panel-npv-functional), component delta-NPV (eq:component-delta-npv), dynamic-policy system (eq:dynamic-policy-system), and policy value (eq:policy-value). It also appears in the new Bellman-style section of the proposal.

Literatures that can close or tighten it. Causal inference and dynamic treatment-regime literatures are most helpful here [12, 14, 23, 25]. They support the distinction between: - the value of a current action under a fixed downstream policy, - and the value of a newly optimized policy.

What the literature actually supports. These references strongly support the requirement that baseline action, downstream policy, horizon, and information set must be explicit if a dynamic incremental value object is to be interpretable.

What remains bank-specific. The bank must still choose the approved baseline semantics, the declared policy-bundle convention, and the exact bridge from one-step incremental cash flow to long-horizon value.

10 Wallet and latent demand identification

Gap in current proposal. This is one of the hardest remaining modeling gaps. The proposal’s cross-sell and cannibalization sections rely on outside wallet, movable wallet, competitor substitution, and promotion-timing objects that are central to bank value but only partially observed.

Where the gap appears. The gap is concentrated in wallet accounting (eq:wallet-accounting), spend-source panel (eq:spend-source-panel), spend-source decomposition (eq:spend-source-decomposition), movable-wallet object (eq:movable-card-wallet), incremental spend-source (eq:incremental-spend-source), new-card lift (eq:new-card-lift), bank-spend lift (eq:bank-spend-lift), cannibalization (eq:cannibalization), and cannibalization-to-NPV (eq:cannibalization-to-npv).

Literatures that can close or tighten it. Payment-choice and instrument-use models already in the bibliography are a first anchor [1, 18, 24]. The causal/uplift literature helps with treatment-effect framing [7, 22, 26, 34]. The new state-space and belief-state references justify modeling outside wallet and movable wallet as latent state rather than observed fact [8, 9, 16, 29]. Banking-specific dynamic control papers such as Trench et al. [32] and So and Thomas [30] add weight that a dynamic value system can legitimately include credit-card profit and customer-state transitions even when not all drivers are observed directly.

What the literature actually supports. These literatures support separating adoption, use, routing, spend source, and policy response. They also support treating wallet-share and substitution objects as latent or partially anchored state components instead of pretending they are directly observed.

What remains bank-specific. No public literature identifies this issuer’s exact competitor-wallet capture, movable-wallet fraction, or promotion-decay profile. Those remain bank-specific latent demand objects to be estimated, bounded, or scenario-tagged with internal anchors.

11 Relationship value integration

Gap in current proposal. The proposal already treats relationship spillovers cautiously, but the dynamic model raises a harder question: when should relationship value be part of the core NPV object, and when should it remain a separately reported spillover?

Where the gap appears. This matters at relationship value (eq:relationship-value), in the valuation-semantics section, and in the allowed-use and evidence-grade sections.

Literatures that can close or tighten it. The current CLV and customer-equity references [3, 6, 27, 33] support the general idea that multi-period customer value can include multiple revenue and cost channels. Banking-specific relationship and value references such as Lewis [21] and Egan et al. [10] can help explain why relationship-level value and deposit franchise value matter in bank economics, even if they do not directly identify card-driven spillovers.

What the literature actually supports. The literature supports reporting relationship value separately or giving it an explicit evidence grade. It supports the idea that relationship economics matter for bank value and customer policy design.

What remains bank-specific. The literature does not justify casually adding relationship spillover to core causal card NPV. Whether and when to include it remains an issuer-specific identification and governance choice.

12 Boundary between fixed-policy valuation and optimization

Gap in current proposal. The proposal now contains dynamic-policy, belief-state, and Bellman language. That strengthens the theoretical backbone, but it also creates a boundary risk: is the proposal still a valuation component under a declared policy, or is it now quietly becoming an optimizer?

Where the gap appears. This gap appears in dynamic-policy system (eq:dynamic-policy-system), policy value (eq:policy-value), the state-space assembly section, the allowed-use taxonomy, and the governance / migration sections.

Literatures that can close or tighten it. The MDP and dynamic treatment-regime references support fixed-policy value and later optimization [23, 25]. The off-policy evaluation and safe-improvement literature [13, 15, 17, 19] supports an offline-first ladder: - fixed-policy simulation, - challenger offline evaluation, - and only then constrained policy improvement.

What the literature actually supports. These literatures support the proposal's current direction if it says clearly: fixed-policy valuation is inside current scope; optimization is a future, governed extension. They support a staged path, not immediate production RL.

What remains bank-specific. The bank must still determine logging adequacy, overlap/support quality, model risk appetite, safe rollout threshold, and whether a valuation component should ever directly feed an optimizing policy layer in production.

13 What the literature can close more fully than bank evidence can

Some gaps are especially amenable to literature support. In particular, the proposal can become much stronger theoretically by citing literatures on:

1. explicit filtration and belief-state sufficiency,
2. dynamic state aggregation and approximation,
3. common-shock macro factorization,
4. the difference between fixed-policy value and optimized-policy value,
5. and conservative offline policy evaluation / safe improvement.

These are places where the literature can do real work for the proposal by clarifying formal language and valid abstraction.

By contrast, some gaps remain stubbornly empirical even after the literature is added:

1. issuer-specific outside-wallet capture,
2. movable-wallet magnitude,

3. relationship spillover causality,
4. exact behavior-policy logging quality,
5. and promotion thresholds for live challenger deployment.

Those should remain explicitly flagged as bank-specific.

14 Writing and citation discipline for the proposal and companion notes

This note recommends the following discipline for the main proposal and the stochastic-process note.

1. If a statement is about why a modeling language is appropriate, cite the formal literature.
2. If a statement is about a dynamic equation already in the proposal, cross-reference the proposal equation directly.
3. If a statement is about a new assembly equation, label it as proposal derivation or assembly derivation.
4. If a statement is about an issuer-specific quantity, say that explicitly and do not let the literature citation imply otherwise.
5. If a claim is only scenario, benchmark, or prior support, label it as such and do not let it masquerade as identification.

The note should therefore be read as a committee-defense crosswalk, not as a claim that the literatures close every gap completely.

15 Residual bank-specific open items

Even after the literature crosswalk, the bank still has to resolve:

1. the exact dynamic state unit and aggregation level by use case,
2. the actual observed feature set and source-system timing contract,
3. the latent-state approximation used in production,
4. the internal anchors for wallet and substitution objects,
5. the relationship-value inclusion rule,
6. the approved valuation-semantics bundle,
7. the behavior-policy logging and overlap diagnostics for offline evaluation,
8. and the governance threshold separating fixed-policy valuation from approved challenger optimization.

These residuals are not a weakness of the note. They are the correct boundary between literature-supported theory and issuer-specific implementation.

16 Conclusion

Yes, there are real literatures that can plug several of the remaining gaps in the current proposal. They are strongest where the proposal needs: - formal dynamic-language support, - explicit filtration and belief-state support, - structured aggregation and common-shock state treatment, - and disciplined boundaries between valuation, policy evaluation, and later optimization.

Those literatures can materially strengthen the proposal. But they do not remove the need for issuer-specific evidence on latent wallet capture, relationship spillovers, dynamic support, and deployment thresholds. The correct defense is therefore not that the literature solves the whole bank problem. It is that the literature sharply improves the theoretical and modeling discipline of the proposal while leaving the genuinely issuer-specific quantities where they belong: in internal data, experiments, quasi-experiments, and governance.

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